III B.Tech – I Semester					
Course Code	**WATER RESOURCES**	L	T	P	C
23A01501	**ENGINEERING**	3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the fundamental concepts of hydrology, including precipitation, evaporation, infiltration, and runoff, and their significancein water resource management.
2. **Analyze** hydrographs, unit hydrographs, and groundwater characteristics for estimating water availability and flood management.
3. **Evaluate** the necessity, importance, and methods of irrigation, along with soil-water-plant relationships and irrigation efficiencies.
4. **Apply** silt theories and principles of canal design to ensure efficient water conveyance andmanagement in irrigation systems.
5. **Assess** the principles of diversion head works, water logging, canal lining, and the stability of hydraulic structures on permeable foundations.

Course Outcomes (COs):

After successful completion of this course, students will be able to:

1. Explain the hydrologic cycle, precipitation types, and measurement techniques for rainfall, evaporation, infiltration, and runoff computation.
2. Analyze hydrographs, unit hydrographs, and groundwater flow parameters for flood estimation and water resource planning.
3. Evaluate irrigation requirements, soil-water-plant relationships, duty, delta, and irrigation efficiencies for sustainable agricultural productivity.
4. Apply silt theories and design principles of irrigation canals to ensure effective water conveyance and prevent water logging.
5. Assess the stability of diversion head works, including weirs and barrages, using Bligh's and Khosla's theories for hydraulic structure design.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO10**	**PO 11**	**PO 12**	**PSO 1**	**PSO 2**
CO -1	**3**	**2**	**2**	**1**	**2**	**-**	**-**	**-**	**-**	**2**	**-**	**2**	**2**	**2**
CO -2	**3**	**3**	**3**	**3**	**3**	**-**	**-**	**-**	**-**	**2**	**-**	**2**	**3**	**2**
CO -3	**3**	**3**	**3**	**3**	**2**	**-**	**-**	**-**	**-**	**2**	**-**	**2**	**3**	**3**
CO -4	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**-**	**2**	**-**	**2**	**3**	**3**
CO -5	**3**	**3**	**3**	**3**	**3**	**2**	**2**	**2**	**-**	**2**	**-**	**2**	**3**	**3**

UNIT - I

INTRODUCTION to HYDROLOGY:

Engineering Hydrology and Its Applications; Hydrologic Cycle; Precipitation- Types and forms, Rainfall Measurement, Types of Rain Gauges, Computation of Average Rainfall Over A Basin, Presentation and Interpretation of Rainfall Data. Evaporation- Factors Affecting Evaporation, Measurement of Evaporation; Infiltration- Factors Affecting Infiltration, Measurement of Infiltration, Infiltration Indices; Run off- Factors Affecting Run- off, Computation of Run-Off; Design Flood; Estimation of Maximum Rate of Run-Off; Separation of Base Flow.

UNIT - II

HYDROGRAPH ANALYSIS:

Hydrograph; Unit Hydrograph- Construction and Limitations of Unit Hydrograph, Application of The Unit Hydrograph to The Construction of A Flood Hydrograph Resulting From Rainfall of Unit Duration; S-Hydrograph.GROUND WATER: Introduction; Aquifer; Aquiclude; Aquifuge; Aquifer Parameters Porosity, Specific Yield, Specific Retention; Divisions of Sub–Surface Water; Water Table;

Types of Aquifers; Storage Coefficient-Coefficient of Permeability and Transmissibility

UNIT – III

IRRIGATION:

Introduction; Necessity and Importance of Irrigation; Advantages and Ill Effects of Irrigation; Types of Irrigation; Methods of Application of Irrigation Water; Quality for Irrigation Water. Duty and Delta; Duty At Various Places; Relation Between Duty and Delta; Factors Affecting Duty; Methods of Improving Duty.

WATER REQUIREMENT of CROPS: Types of Soils, Indian Agricultural Soils, Preparation of Land for Irrigation; Soil Fertility; Soil-Water-Plant Relationship; Vertical Distribution of Soil Moisture; Soil Moisture Tension; Soil Moisture Stress; Various Soil Moisture Constants; Limiting Soil Moisture Conditions; Depth and Frequency of Irrigation; Gross Command Area; CulturableCommand Area; CulturableCultivated and Uncultivated Area; KorDepth andKorPeriod; Crop Seasons and Crop Rotation; Irrigation Efficiencies; Determination of Irrigation Requirements of Crops; Assessment of Irrigation Water. Consumptive Use of Water-Factors Affecting Consumptive Use, Direct Measurement and Determination By Use of Equations (Theory Only)

UNIT – IV

CHANNELS – SILT THEORIES:

Classification; Canal Alignment; Inundation Canals; Cross–Section of An Irrigation Channel; Balancing Depth; Borrow Pit; Spoil Bank; Land Width; Silt Theories–Kennedy's Theory, Kennedy's Method of Channel Design; Drawbacks in Kennedy's Theory; Lacey's Regime Theory- Lacey's Theory Applied to Channel Design; Defects in Lacey's Theory; Comparison ofKennedy's andLacey's Theory.

WATER LOGGING and CANAL LINING: Water Logging; Effects of Water Logging; Causes of Water Logging; Remedial Measures; Saline and Alkaline Soils and their Reclamation; Losses in Canal; Lining of Irrigation Channels – Necessity, Advantages and Disadvantages; Types of Lining; Design of Lined Canal.

UNIT – V

DIVERSION HEAD WORKS:

Types of Diversion Head Works; Diversion and Storage Head Works; Weirs and Barrages; Layouts of Diversion Head Works; Components; Causes and Failure of Hydraulic Structures On Permeable Foundations; Bligh's Creep Theory; Khosla's Theory; Determination of Uplift Pressure, Impervious Floors Using Bligh's and Khosla's Theory; Exit Gradient.

TEXT BOOKS:

1. Irrigation and Water Power Engineering By Punmia&Lal, Laxmi Publications Pvt. Ltd., New Delhi 17th Edition 2021
2. Engineering Hydrology By K. Subramanya, The Tata McGraw Hill Company, Delhi 5th Edition 2020

REFRENCE BOOKS:

1. Irrigation Engineering and Hydraulic Structures By S. K. Garg; Khanna Publishers, Delhi 36th Edition
2. Engineering Hydrology By Jayarami Reddy, Laxmi Publications Pvt. Ltd., New Delhi 3rd Edition 2016
3. Irrigation and Water Resources & Water Power By P.N.Modi, Standard Book House 6th Edition 2020

Online Learning Resources:

https://nptel.ac.in/courses/105101214